M15: Final Group Project Report
Identification of Open Reading Frames Using a Python Based ORF Finder

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Project Summary

In this project, we created a Python program that detects Open Reading Frames (ORFs) in DNA sequences provided in FASTA format. An ORF is a sequence that begins with a start codon (ATG) and ends with a stop codon (TAA, TAG, or TGA). To ensure complete detection of all potential coding regions, the program examines all six possible reading frames, including three forward frames and three reverse complement frames. The program processes each sequence by scanning codons in three base steps, identifying start and stop codons, calculating the correct genomic position (positive for forward frames, negative for reverse frames), and reporting ORFs that meet the user specified minimum length requirement.

Division of Labor

The project was divided into three major components, allowing each group member to focus on a specific part of the program:

- Anoushka Prabhu worked on the input parser module. She added functions for reading FASTA files, handling multiple sequences, removing whitespace, and ensuring consistent formatting of DNA sequences.
- Ashwarya Sharma was in charge of ORF detection. She implemented code to generate and analyze the reverse complement sequence and then scan all reading frames for the specific start and stop codons.
- Pratham Patel handled the output formatting. He created functions that convert DNA sequences into codons, limit output to 15 codons per line, and display results in proper FASTA format with headers containing frame number, position, and length.

Following the completion of individual modules, we collaborated to integrate the components and ensure function compatibility through Canvas discussion as well as in-person meetings. To ensure everything integrated smoothly, our group worked collaboratively, communicating and reviewing each other's code on a regular basis.

Challenges Faced

One of the most difficult challenges was accurately implementing ORF detection across all six reading frames. We also ensured that frame numbering followed the required convention of frames 1-3 for the forward strand and 4-6 for the reverse strand. In the first draft of code, the sequence only searched in the three forward frames and completely missed potential ORFs on the reverse strand. This made it important to make sure that the reverse complement strand was properly created and scanned.

Calculating the correct genomic position for ORFs on the reverse strand (frames 4-6) also required careful attention, as positions needed to be reported as negative numbers representing distance from the right end of the sequence

We also encountered issues with input handling and the output of the code. Initially, the program asked the user for the minimum ORF length before determining whether the FASTA file existed, resulting in unnecessary prompts when incorrect file names were entered. This was fixed by restructuring the program so that the minimum length is only requested after the file has been successfully loaded. Similarly, removing unnecessary print statements and ensuring each ORF was displayed properly was necessary to make sure the output matched FASTA formatting.

Finally, there were some discrepancies and inconsistencies in the code since each person developed their part independently. This resulted in integration issues when combined at first. We addressed this issue by standardizing variable names and ensuring all functions used consistent input and output formats.

Future Improvements

There are several ways to improve the program in the future. One improvement is to handle ambiguous nucleotide characters more effectively, either by removing invalid characters during parsing or warning users about their presence. Another useful addition would be to allow users to save output to a file rather than just displaying it in the console. Expanding the user interface and adding command line options would also make the program more usable for research projects. These changes would make the program more robust for actual sequencing data, where raw files often contain ambiguous bases and users need to archive results for later analysis.

The program could also be expanded to include alternative start or stop codons based on the organism of interest. For example, researchers working with multiple organisms would benefit if codon sets could be selected directly from the input rather than having to manually change them in the code for each experiment. Additionally, the code could be modified to use multi-threading, allowing parallel processing of multiple sequences or reading frames to improve performance on large genomic datasets.

Conclusion

Overall, our team worked in close collaboration to create an ORF Finder program that identifies open reading frames in DNA sequences across multiple sequences successfully. Throughout the project, we ensured that all functions used consistent input and output formats. The program meets all assignment specifications, including proper FASTA output, correct frame numbering, accurate position calculation (positive for forward frames, negative for reverse frames), and codon formatting with 15 codons per line. This project also helped us improve our teamwork and communication skills as we worked on a shared coding task.